

Gas Detection Circuit using LCD Display

A real-time, standalone gas-leakage monitoring system built with an Arduino Uno, an MQ-series gas sensor, and a 16x2 LCD readout.

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What This Project Does

This circuit continuously samples the surrounding air with a gas sensor and displays a live, human-readable status on an onboard LCD screen -- no computer or app required.

1 Continuous Sensing

The MQ gas sensor's analog output is sampled on Arduino pin A0 in a constant loop.

2 Instant Local Feedback

Readings and alerts are shown directly on a 16x2 character LCD -- no phone or PC needed.

3 Adjustable Contrast

A 250kΩ potentiometer tunes LCD contrast for clear readability in any lighting.

4 Simple, Low-Cost Build

Only six components: Arduino Uno, gas sensor, LCD, potentiometer, and two resistors.

MQ

GAS SENSOR

Analog resistance changes in the presence of combustible or toxic gases, producing a voltage read on pin A0.

Output → Arduino A0

Bill of Materials

Six components, one breadboard -- everything the circuit needs to run.

Ref.	Component	Qty	Role in Circuit
U1	Arduino Uno R3	1	Microcontroller -- reads the sensor and drives the LCD
GAS1	MQ Gas Sensor	1	Detects combustible / toxic gas concentration in air
U2	16x2 LCD Display	1	Shows live gas readings and status messages
RPOT2	250 k Ω Potentiometer	1	Adjusts LCD contrast (connected to VO)
R1	1 k Ω Resistor	1	Current-limiting resistor in the sensor/LED stage
R2	330 Ω Resistor	1	Current-limiting resistor for the sensor's heater/LED

How It Works

1

Sense

The MQ gas sensor's sensing element changes resistance as gas concentration in the air rises, producing a variable analog voltage.

2

Read

The Arduino Uno continuously samples that voltage on analog pin A0 and converts it to a digital value (0-1023) via its ADC.

3

Process

The sketch compares the reading against a threshold to decide whether air quality is normal or a gas leak is likely.

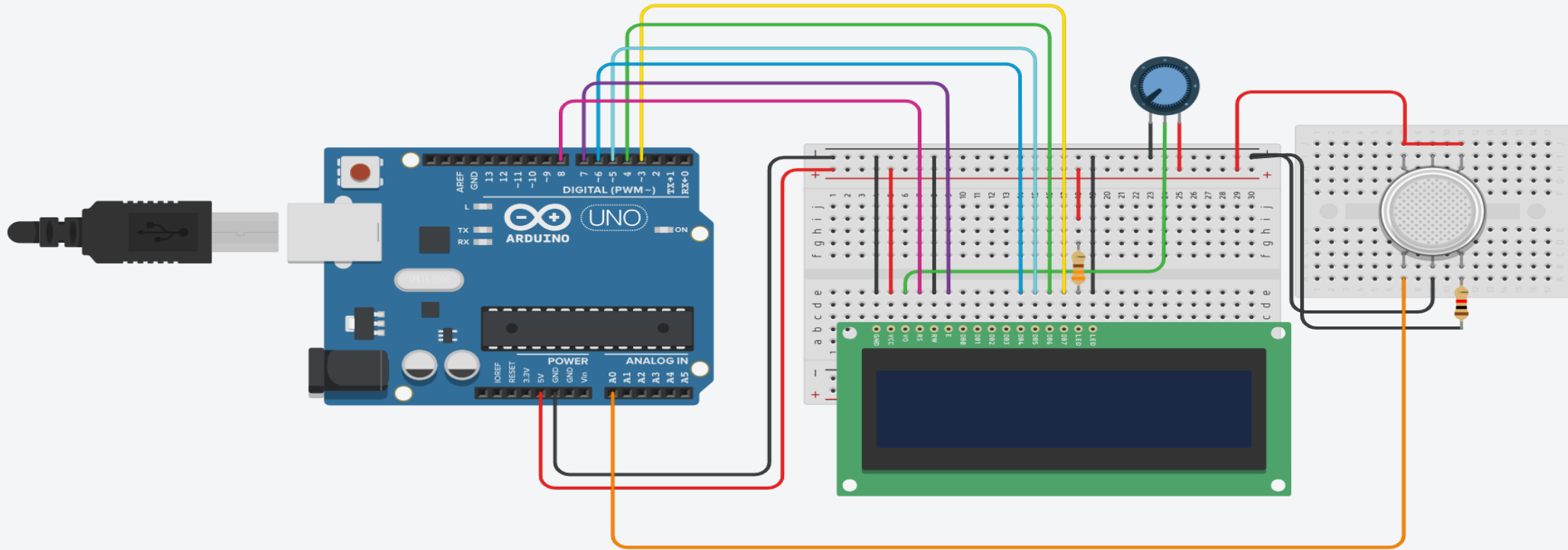
4

Display

The result is written to the 16x2 LCD (driven over 6 digital pins) in real time, giving an immediate on-site readout.

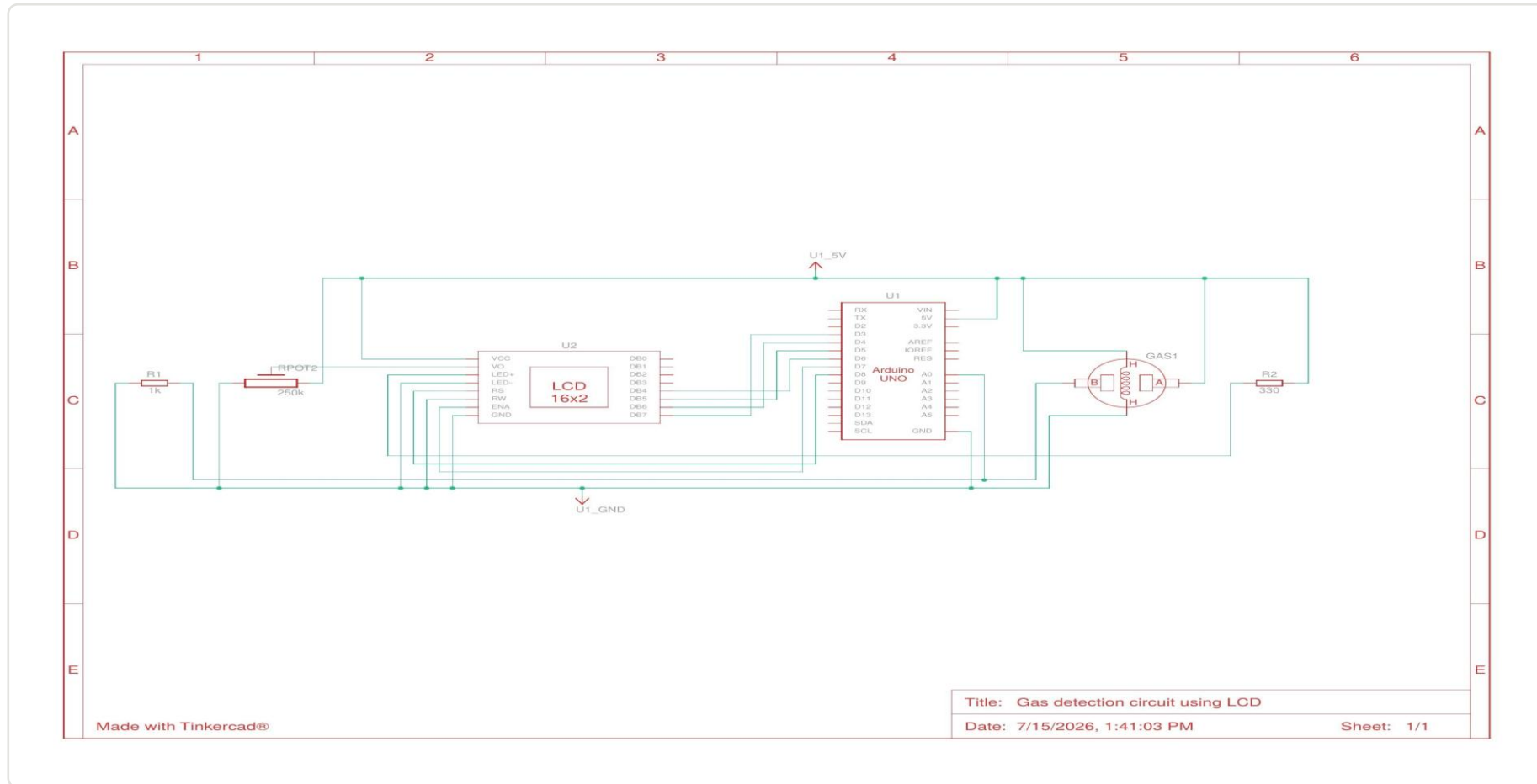
Breadboard Circuit Diagram

Full wiring as simulated in Tinkercad



Schematic Diagram

Electrical connections and component symbols



Signal Connections

Functional pin mapping between the Arduino Uno and each peripheral

Arduino Uno	Connects To	Purpose
A0 (Analog In)	Gas Sensor (GAS1) Output	Reads variable gas-concentration voltage
D2 - D8 (Digital)	LCD 16x2 (RS, E, D4-D7)	Drives the LCD in 4-bit mode
5V / GND	LCD VCC/GND, Sensor, Potentiometer	Shared power and ground rails
Potentiometer Wiper	LCD VO (Contrast Pin)	Sets LCD display contrast
R1 / R2	Gas Sensor Load Path	Limit current through the sensor stage

Note: LCD data lines are driven in 4-bit mode, so only 6 of the 8 data pins (D4-D7 plus RS and Enable) are wired.

Applications



Home Safety

Early warning for LPG or smoke leaks in kitchens and utility rooms.



Industrial Monitoring

Continuous air-quality checks near storage tanks and process lines.



Laboratories

Localized alerts where solvent or gas fumes pose a health risk.



Parking Garages

Detects vehicle exhaust build-up in enclosed spaces.



Agricultural Storage

Monitors silos and grain stores for hazardous gas accumulation.



Educational Demos

A compact, low-cost platform for teaching embedded sensing.

Future Enhancements

- 1 Add a buzzer / relay output to trigger an audible alarm or shut off a valve automatically.
- 2 Log readings over time with an SD card or send them to the cloud (Wi-Fi / IoT module) for remote monitoring.
- 3 Calibrate against multiple MQ sensor variants to detect specific gases (LPG, CO, smoke, methane).
- 4 Add a mobile app or SMS alert path for notifications when no one is near the display.

Thank You